

A positive-conservative Fourier optimization method for ground states of Bose–Einstein condensates with higher-order interactions

Bo Lin,^{*} Xinran Ruan[†]

Abstract

We develop a positive-conservative Fourier optimization (PCFO) method for computing ground states of Bose–Einstein condensates with higher-order interactions. The ground-state problem admits a convex density formulation, but the singular behavior near zero density poses difficulties for high-accuracy computation. We introduce convex regularizations of the density energy and establish their Γ -convergence on the computational domain as the regularization parameters vanish. The regularized problem is discretized by a Fourier pseudospectral method with mass conservation and nodal positivity. For the resulting constrained optimization problem, we use an accelerated first-order method for the main energy reduction, followed by a Newton refinement to reduce the remaining optimality residual. Numerical experiments show spectral-type convergence for regularized problems, quantify the effects of the regularization parameters on accuracy and spatial resolution, and demonstrate the effectiveness of the two-stage optimization method.

Key words. Bose–Einstein condensate; higher-order interaction; density formulation; convex optimization; Fourier pseudospectral method; positivity preserving.

2020 MSC. 35Q55, 65N35, 65K10, 90C25.

1 Introduction

Bose–Einstein condensation, first realized experimentally in dilute atomic gases in 1995 [2], provides a fundamental setting for the study of macroscopic quantum phenomena. In the dilute and low-temperature regime, Bose–Einstein condensates are commonly described by the Gross–Pitaevskii equation [5], in which two-body interactions are modeled by a zero-range contact potential. Finite-range and effective-range corrections arise beyond this approximation [17, 20] and may become more relevant under strong confinement or near narrow Feshbach resonances [16, 26, 28]. One such correction leads to a modified Gross–Pitaevskii model with a higher-order interaction involving the gradient of the density.

We consider the modified Gross–Pitaevskii model with higher-order interactions (HOI), whose energy is

$$E(\phi) = \int_{\mathbb{R}^d} \left[\frac{1}{2} |\nabla \phi|^2 + V(x) |\phi|^2 + \frac{\beta}{2} |\phi|^4 + \frac{\delta}{2} |\nabla(|\phi|^2)|^2 \right] dx, \quad (1.1)$$

^{*}Beijing Huairou Laboratory, Beijing 101400, People’s Republic of China

[†](Corresponding author) School of Mathematical Sciences, Capital Normal University, Beijing 100048, People’s Republic of China (xinran.ruan@cnu.edu.cn).

subject to

$$\|\phi\|_{L^2}^2 = 1. \quad (1.2)$$

Here ϕ is the macroscopic wave function of the condensate, $V(x) \geq 0$ is a confining potential, $\beta \geq 0$ is the repulsive contact-interaction strength, and $\delta > 0$ is the higher-order interaction strength. The competition between the contact and higher-order interactions can affect ground-state structures and their asymptotic regimes [6, 23]. Under strong confinement, the higher-order interaction also modifies the effective lower-dimensional mean-field models [23].

As shown in [6], a ground state may be chosen real and nonnegative. This motivates working with the density $\rho = |\phi|^2$, which gives the equivalent formulation

$$\mathcal{E}_0(\rho) = \int_{\mathbb{R}^d} \left[\frac{|\nabla \rho|^2}{8\rho} + V(x)\rho + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \quad (1.3)$$

over

$$\mathcal{A} = \left\{ \rho \in H^1(\mathbb{R}^d) : \rho \geq 0, \quad \int_{\mathbb{R}^d} \rho \, dx = 1, \quad \int_{\mathbb{R}^d} V\rho \, dx < \infty \right\}. \quad (1.4)$$

The quotient $|\nabla \rho|^2/(8\rho)$ is interpreted in the extended sense defined in Section 2. For $\beta \geq 0$, $\mathcal{E}_0$ is strictly convex on $\mathcal{A}$. This convexity is useful for ground-state computation, although the kinetic term remains singular at zero density.

Ground-state computation for the Gross–Pitaevskii equation has been studied extensively. Early approaches include direct energy minimization and normalized gradient-flow methods [8, 10], together with spectrally accurate discretizations [7]. For strongly nonlinear or rotating condensates, Sobolev-gradient methods, preconditioned spectral solvers, nonlinear conjugate-gradient methods, regularized Newton methods, and Riemannian optimization have been developed [3, 4, 18, 19, 27]. More recent work includes normalized Sobolev-gradient flows and nonlinear eigenvalue iterations with convergence analysis [1, 15, 21]. Fourier pseudospectral discretizations combined with manifold optimization have also been applied to more general spinor condensates [24].

For the HOI model, a normalized gradient-flow method with an attractive–repulsive splitting was developed in [22]. W. Bao and X. Ruan [9] subsequently formulated the ground-state problem as a convex minimization problem in the density variable, using a shifted kinetic regularization, a second-order finite-difference discretization, and an accelerated projected-gradient method, with a square-root regularization of the potential considered numerically.

To enable spectral spatial accuracy within the density formulation, we further develop the framework of [9] by improving both the spatial discretization and the optimization solver. The second-order finite-difference discretization is replaced by a Fourier pseudospectral method with exact mass conservation and nodal positivity. The potential term is incorporated into the proximal step of the accelerated first-order method, followed by a Newton refinement to reduce the optimization error so that it does not limit the spatial discretization accuracy. We also extend the shifted kinetic regularization to a class of convexity-preserving denominators and treat the potential regularization with an independent scale. The resulting regularized energies are shown to Γ -converge on the computational domain as the regularization parameters vanish. The spatial accuracy, regularization effects, and solver performance are then examined numerically.

The rest of the paper is organized as follows. Section 2 introduces the convex regularizations and establishes their variational convergence. Section 3 presents the Fourier pseudospectral discretization and the projection preserving positivity and mass. Section 4 develops the potential-proximal FISTA method and the Newton refinement. Section 5 reports the numerical results, and Section 6 concludes the paper.

2 Density formulation and convex regularization

In this section, we study the convex structure of the density energy, introduce convex regularizations of the kinetic and potential terms, and establish variational convergence of the regularized energies.

2.1 Convex regularization of the kinetic and potential terms

Kinetic regularization. We extend the density kinetic term to possible vacuum regions. For $s \geq 0$ and $q \in \mathbb{R}^d$, let

$$F_0(s, q) = \begin{cases} \frac{|q|^2}{8s}, & s > 0, \\ 0, & s = 0, \quad q = 0, \\ +\infty, & s = 0, \quad q \neq 0. \end{cases} \quad (2.1)$$

With this convention, the density energy in (1.3) can be written as

$$\mathcal{E}_0(\rho) = \int_{\mathbb{R}^d} \left[F_0(\rho, \nabla \rho) + V\rho + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \quad \rho \in \mathcal{A}. \quad (2.2)$$

The density energy has the following convexity property.

Proposition 2.1 (Convexity of the density energy). *Assume $\beta \geq 0$ and $\delta > 0$. Then $\mathcal{E}_0$ is strictly convex on $\mathcal{A}$ and hence has at most one minimizer.*

Proof. The map $(s, q) \mapsto |q|^2/s$ is the perspective of the convex quadratic function $q \mapsto |q|^2$, and (2.1) is its lower-semicontinuous convex extension to $s = 0$. Hence the kinetic term is convex. The potential term is linear, and the β term is convex for $\beta \geq 0$.

The δ term gives strict convexity on $\mathcal{A}$. Indeed, equality in its convexity inequality for two densities in $\mathcal{A}$ implies that their gradients coincide almost everywhere. Their difference is therefore constant, and integrability on $\mathbb{R}^d$ implies that this constant is zero. $\square$

To remove the singularity at zero density while preserving the convex structure, we regularize the kinetic denominator. For $s \geq 0$ and $q \in \mathbb{R}^d$, set

$$F_\varepsilon(s, q) = \frac{|q|^2}{8r_\varepsilon(s)}, \quad (2.3)$$

where r_ε is a positive regularization of the denominator.

Definition 2.2 (Admissible denominator). A family $\{r_\varepsilon\}_{\varepsilon>0}$ is called admissible if, for every $\varepsilon > 0$,

$$r_\varepsilon \in C^1([0, \infty)), \quad r_\varepsilon > 0, \quad r'_\varepsilon \geq 0, \quad r_\varepsilon \text{ is concave}, \quad (2.4)$$

and, for every $s \geq 0$,

$$s \leq r_{\varepsilon_1}(s) \leq r_{\varepsilon_2}(s) \quad \text{for } 0 < \varepsilon_1 \leq \varepsilon_2, \quad r_\varepsilon(s) \rightarrow s \quad \text{as } \varepsilon \rightarrow 0^+. \quad (2.5)$$

The concavity of the denominator preserves the convexity of the regularized kinetic term.

Lemma 2.3 (Convexity of the regularized kinetic integrand). *Let r_ε be admissible in the sense of Definition 2.2. Then*

$$F_\varepsilon(s, q) = \frac{|q|^2}{8r_\varepsilon(s)}$$

is jointly convex in $(s, q) \in [0, \infty) \times \mathbb{R}^d$.

Proof. Let $0 \leq \theta \leq 1$ and set

$$s_\theta = \theta s_1 + (1 - \theta)s_2, \quad q_\theta = \theta q_1 + (1 - \theta)q_2.$$

By the concavity of r_ε ,

$$r_\varepsilon(s_\theta) \geq \theta r_\varepsilon(s_1) + (1 - \theta)r_\varepsilon(s_2).$$

Hence

$$F_\varepsilon(s_\theta, q_\theta) \leq \frac{|q_\theta|^2}{8[\theta r_\varepsilon(s_1) + (1 - \theta)r_\varepsilon(s_2)]}.$$

The weighted Cauchy–Schwarz inequality gives

$$\frac{|q_\theta|^2}{\theta r_\varepsilon(s_1) + (1 - \theta)r_\varepsilon(s_2)} \leq \theta \frac{|q_1|^2}{r_\varepsilon(s_1)} + (1 - \theta) \frac{|q_2|^2}{r_\varepsilon(s_2)},$$

which proves the result. $\square$

The simplest example is the shifted denominator introduced in [9],

$$r_\varepsilon(s) = s + \varepsilon. \quad (2.6)$$

For comparison, we also consider the family

$$r_{\varepsilon,m}(s) = \begin{cases} s + \varepsilon - \frac{\varepsilon}{m+1} \left(1 - \frac{s}{\varepsilon}\right)^{m+1}, & 0 \leq s \leq \varepsilon, \\ s + \varepsilon, & s > \varepsilon, \end{cases} \quad m = 1, 2. \quad (2.7)$$

These functions belong to $C^m([0, \infty))$ and satisfy Definition 2.2.

Potential regularization. The kinetic regularization removes the singularity at $\rho = 0$, but the derivative of the linear potential term $V\rho$ at zero density is V . In the discrete problem, this can activate the positivity constraint and reduce the regularity relevant to the Fourier approximation. We therefore regularize the potential term near zero density by replacing ρ with $p_\sigma(\rho)$, where

$$p_\sigma(s) = \sqrt{s^2 + \sigma^2} - \sigma, \quad s \geq 0. \quad (2.8)$$

A square-root smoothing of this type was considered in [9] as a numerical enhancement. Here σ is treated as an independent regularization scale and is analyzed together with the kinetic regularization. We have

$$p_\sigma(0) = p'_\sigma(0) = 0, \quad p''_\sigma(s) = \frac{\sigma^2}{(s^2 + \sigma^2)^{3/2}} \geq 0. \quad (2.9)$$

Thus Vp_σ is convex for $V \geq 0$. The condition $p'_\sigma(0) = 0$ removes the nonzero derivative of the potential term at zero density. Moreover, $p_\sigma(s) \rightarrow s$ as $\sigma \rightarrow 0^+$. For the convergence analysis below, it is sufficient to take $\sigma = \sigma(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0^+$.

Combining the two regularizations, we obtain

$$\mathcal{E}_{\varepsilon,\sigma}(\rho) = \int_{\mathbb{R}^d} \left[F_\varepsilon(\rho, \nabla \rho) + V(x)p_\sigma(\rho) + \frac{\beta}{2}\rho^2 + \frac{\delta}{2}|\nabla \rho|^2 \right] dx, \quad \rho \in \mathcal{A}. \quad (2.10)$$

The convex structure of the original density problem is retained by the regularized energy.

Proposition 2.4 (Convexity of the regularized energy). *Assume that r_ε is admissible in the sense of Definition 2.2, $V \geq 0$, $\beta \geq 0$, $\delta > 0$, and $\sigma > 0$. Then $\mathcal{E}_{\varepsilon,\sigma}$ (2.10) is strictly convex on $\mathcal{A}$ (1.4). If $\beta > 0$, it is strongly convex with respect to the L^2 norm.*

Proof. By Lemma 2.3, the kinetic term is convex. The potential term is convex because p_σ is convex and $V \geq 0$, while the remaining terms are convex. Strict convexity follows from the δ term as in Proposition 2.1. If $\beta > 0$, the term $\frac{\beta}{2}\|\rho\|_{L^2}^2$ gives strong convexity in L^2 . $\square$

2.2 Γ -convergence of the regularized energies

To formulate the Fourier discretization on a bounded domain, we introduce the periodic box

$$\mathcal{D}_L = [-L, L]^d.$$

On $\mathcal{D}_L$, we study the Γ -convergence of the regularized energies as the regularization parameters vanish, with respect to the strong $L^2(\mathcal{D}_L)$ topology [12].

Let $H_{\text{p}}^1(\mathcal{D}_L)$ denote the periodic Sobolev space on $\mathcal{D}_L$. Define

$$\mathcal{A}_L = \left\{ \rho \in H_{\text{p}}^1(\mathcal{D}_L) : \rho \geq 0 \text{ a.e., } \int_{\mathcal{D}_L} \rho \, dx = 1 \right\}. \quad (2.11)$$

For $\rho \in \mathcal{A}_L$, define

$$\begin{aligned} \mathcal{E}_{\varepsilon, \sigma, L}(\rho) &= \int_{\mathcal{D}_L} \left[F_{\varepsilon}(\rho, \nabla \rho) + V p_{\sigma}(\rho) + \frac{\beta}{2} \rho^2 + \frac{\delta}{2} |\nabla \rho|^2 \right] dx, \\ \mathcal{E}_{0, L}(\rho) &:= \mathcal{E}_{0, 0, L}(\rho) = \int_{\mathcal{D}_L} \left[F_0(\rho, \nabla \rho) + V \rho + \frac{\beta}{2} \rho^2 + \frac{\delta}{2} |\nabla \rho|^2 \right] dx. \end{aligned} \quad (2.12)$$

The potential regularization is uniformly small on $\mathcal{D}_L$. Since

$$0 \leq s - p_{\sigma}(s) \leq \sigma, \quad s \geq 0, \quad (2.13)$$

we have

$$\left| \int_{\mathcal{D}_L} V [p_{\sigma}(\rho) - \rho] \, dx \right| \leq \sigma \int_{\mathcal{D}_L} V \, dx.$$

Thus the potential contribution is a uniformly vanishing perturbation as $\sigma \rightarrow 0$, and the main convergence argument can be carried out for the kinetic regularization.

Theorem 2.5 (Γ -convergence of the joint regularization). *Let $V \in C(\mathcal{D}_L)$ with $V \geq 0$, $\beta \geq 0$, and $\delta > 0$. Let $\{r_{\varepsilon}\}$ be admissible in the sense of Definition 2.2, and let $\sigma(\varepsilon) > 0$ satisfy $\sigma(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0^+$. Extend $\mathcal{E}_{\varepsilon, \sigma(\varepsilon), L}$ and $\mathcal{E}_{0, L}$ to $L^2(\mathcal{D}_L)$ by setting them equal to $+\infty$ outside $\mathcal{A}_L$. Then*

$$\mathcal{E}_{\varepsilon, \sigma(\varepsilon), L} \xrightarrow{\Gamma} \mathcal{E}_{0, L} \quad \text{in the strong } L^2(\mathcal{D}_L) \text{ topology.}$$

Moreover, the family $\{\mathcal{E}_{\varepsilon, \sigma(\varepsilon), L}\}_{\varepsilon > 0}$ is equicoercive in $L^2(\mathcal{D}_L)$.

Proof. We first consider the kinetic regularization alone and keep the potential term unchanged. Define

$$\tilde{\mathcal{E}}_{\varepsilon, L}(\rho) = \int_{\mathcal{D}_L} \left[F_{\varepsilon}(\rho, \nabla \rho) + V \rho + \frac{\beta}{2} \rho^2 + \frac{\delta}{2} |\nabla \rho|^2 \right] dx.$$

We first prove the liminf inequality. Let $\varepsilon_n \rightarrow 0$ and $\rho_n \rightarrow \rho$ strongly in $L^2(\mathcal{D}_L)$. If

$$\liminf_{n \rightarrow \infty} \tilde{\mathcal{E}}_{\varepsilon_n, L}(\rho_n) = +\infty,$$

there is nothing to prove. Otherwise, after passing to a subsequence, we may assume that the energies are uniformly bounded. Since $\delta > 0$, the gradient term together with the fixed mass gives a uniform $H^1(\mathcal{D}_L)$ bound. Hence, after extracting a further subsequence,

$$\rho_n \rightharpoonup \rho \quad \text{weakly in } H^1(\mathcal{D}_L).$$

Fix $\bar{\varepsilon} > 0$. For all sufficiently large n , $\varepsilon_n \leq \bar{\varepsilon}$. The ordering in (2.5) therefore gives

$$F_{\varepsilon_n}(s, q) \geq F_{\bar{\varepsilon}}(s, q).$$

For fixed $\bar{\varepsilon}$, the integrand $F_{\bar{\varepsilon}}$ is continuous and jointly convex in (s, q) . The associated integral functional is therefore weakly lower semicontinuous on $H^1(\mathcal{D}_L)$. Together with the remaining terms, this yields

$$\liminf_{n \rightarrow \infty} \tilde{\mathcal{E}}_{\varepsilon_n, L}(\rho_n) \geq \tilde{\mathcal{E}}_{\bar{\varepsilon}, L}(\rho).$$

Finally, let $\bar{\varepsilon} \downarrow 0$. Since $r_{\bar{\varepsilon}}(s) \downarrow s$, the nonnegative kinetic integrands increase pointwise to F_0 . The monotone convergence theorem then gives

$$\liminf_{n \rightarrow \infty} \tilde{\mathcal{E}}_{\varepsilon_n, L}(\rho_n) \geq \mathcal{E}_{0, L}(\rho).$$

For the recovery sequence, let $\rho \in \mathcal{A}_L$ satisfy $\mathcal{E}_{0, L}(\rho) < \infty$ and take the constant sequence $\rho_\varepsilon = \rho$. The same monotone convergence argument gives

$$\tilde{\mathcal{E}}_{\varepsilon, L}(\rho) \longrightarrow \mathcal{E}_{0, L}(\rho) \quad \text{as } \varepsilon \rightarrow 0^+.$$

Hence

$$\tilde{\mathcal{E}}_{\varepsilon, L} \xrightarrow{\Gamma} \mathcal{E}_{0, L} \quad \text{in strong } L^2(\mathcal{D}_L).$$

It remains to restore the potential regularization. By (2.13),

$$\sup_{\rho \in \mathcal{A}_L} \left| \mathcal{E}_{\varepsilon, \sigma(\varepsilon), L}(\rho) - \tilde{\mathcal{E}}_{\varepsilon, L}(\rho) \right| \leq \sigma(\varepsilon) \int_{\mathcal{D}_L} V \, dx \longrightarrow 0.$$

Thus the potential smoothing is a uniformly vanishing perturbation and does not change the Γ -limit.

The equicoercivity follows from the δ term and the fixed mass, which give a uniform $H^1(\mathcal{D}_L)$ bound, together with the compact embedding $H^1(\mathcal{D}_L) \hookrightarrow L^2(\mathcal{D}_L)$. $\square$

The Γ -convergence result gives the corresponding convergence of minimum values and minimizers.

Corollary 2.6 (Convergence of minimizers). *Under the assumptions of Theorem 2.5, let $\rho_{\varepsilon, L}$ and $\rho_{0, L}$ denote the minimizers of $\mathcal{E}_{\varepsilon, \sigma(\varepsilon), L}$ and $\mathcal{E}_{0, L}$ on $\mathcal{A}_L$, respectively. Then*

$$\min_{\rho \in \mathcal{A}_L} \mathcal{E}_{\varepsilon, \sigma(\varepsilon), L}(\rho) \longrightarrow \min_{\rho \in \mathcal{A}_L} \mathcal{E}_{0, L}(\rho), \quad \rho_{\varepsilon, L} \longrightarrow \rho_{0, L} \quad \text{strongly in } L^2(\mathcal{D}_L). \quad (2.14)$$

Proof. We first show that the minimizers exist and are unique. Existence follows from the direct method. Indeed, the δ term together with the fixed mass gives the required $H^1(\mathcal{D}_L)$ bound for a minimizing sequence, while weak lower semicontinuity and compactness on the bounded periodic domain yield a minimizer in $\mathcal{A}_L$.

For uniqueness, the convexity arguments of Propositions 2.1 and 2.4 remain valid on $\mathcal{D}_L$. Since all terms in the energy are convex, equality in the convexity inequality implies equality for the δ term. Hence two minimizers must have the same gradient and therefore differ by a constant. In the whole-space case this constant is excluded by integrability, whereas on the periodic domain it is excluded by the fixed-mass constraint. Thus the constant is zero and the minimizer is unique.

Theorem 2.5 and equicoercivity then give the convergence of the minimum values and the subsequential convergence of $\rho_{\varepsilon, L}$. Every convergent subsequence has a minimizer of $\mathcal{E}_{0, L}$ as its limit. Since this minimizer is unique, all such limits coincide with $\rho_{0, L}$, and therefore the whole family converges. $\square$

3 Fourier pseudospectral discretization

We discretize the regularized density problem on a periodic computational domain by a Fourier pseudospectral method. We introduce the discrete energy and its derivatives, together with the positive mass-conservative projection used in the constrained optimization.

We present the one-dimensional formulation on $\mathcal{D}_L = [-L, L]$. The multidimensional case follows by the standard tensor-product extension. Let N be even, $x_j = -L + jh$, $j = 0, \dots, N-1$, and $h = 2L/N$. For $u, v \in \mathbb{R}^N$, define the discrete inner product and norm by

$$\langle u, v \rangle_h = h \sum_{j=0}^{N-1} u_j v_j, \quad \|u\|_h^2 = \langle u, u \rangle_h. \quad (3.1)$$

3.1 Discrete energy, gradient and Hessian structure

Let D_N denote the real Fourier pseudospectral first-derivative operator. For even N , we set the derivative of the Fourier mode $k = N/2$ to zero, so that D_N maps real grid vectors to real grid vectors and satisfies

$$D_N^T = -D_N \quad (3.2)$$

with respect to the inner product (3.1). The action of D_N is evaluated by FFTs.

For a grid density ρ , let $q = D_N \rho$. The discrete regularized energy is

$$E_{\varepsilon, \sigma, N}(\rho) = h \sum_{j=0}^{N-1} \left[\frac{q_j^2}{8r_\varepsilon(\rho_j)} + V_j p_\sigma(\rho_j) + \frac{\beta}{2} \rho_j^2 + \frac{\delta}{2} q_j^2 \right], \quad (3.3)$$

subject to the discrete positivity and mass constraints

$$\mathcal{C}_N = \left\{ \rho \in \mathbb{R}^N : \rho_j \geq 0, \quad h \sum_{j=0}^{N-1} \rho_j = 1 \right\}. \quad (3.4)$$

Proposition 3.1 shows that the discrete energy (3.3) retains the convexity of the regularized problem.

Proposition 3.1 (Discrete convexity). *Let r_ε be admissible in the sense of Definition 2.2. Let $\sigma > 0$ and assume $V_j \geq 0$, $\beta \geq 0$, and $\delta > 0$. Then $E_{\varepsilon, \sigma, N}$ is convex on $\mathcal{C}_N$ and admits a minimizer. If $\beta > 0$, it is strongly convex with respect to $\|\cdot\|_h$, and the minimizer is unique.*

Proof. By Lemma 2.3 and the linearity of

$$\rho \mapsto (\rho_j, (D_N \rho)_j),$$

the discrete kinetic term is convex. The remaining terms inherit the convexity properties established in Proposition 2.4. Since $\mathcal{C}_N$ is compact and $E_{\varepsilon, \sigma, N}$ is continuous, a minimizer exists. If $\beta > 0$, the term $\frac{\beta}{2} \|\rho\|_h^2$ gives strong convexity and hence uniqueness. $\square$

Differentiating (3.3) gives the discrete gradient. Let $q = D_N \rho$. With all nonlinear operations understood pointwise, the gradient with respect to the inner product (3.1) is characterized by

$$G_{\varepsilon, \sigma, N}(\rho) = \frac{1}{4} D_N^T \left(\frac{q}{r_\varepsilon(\rho)} \right) - \frac{1}{8} \frac{q^2 r'_\varepsilon(\rho)}{r_\varepsilon(\rho)^2} + V p'_\sigma(\rho) + \beta \rho + \delta D_N^T D_N \rho, \quad (3.5)$$

$$\left. \frac{d}{dt} E_{\varepsilon, \sigma, N}(\rho + t\eta) \right|_{t=0} = \langle G_{\varepsilon, \sigma, N}(\rho), \eta \rangle_h.$$

At points where r_ε is twice differentiable, define

$$B = D_N - \text{diag} \left(\frac{q r'_\varepsilon(\rho)}{r_\varepsilon(\rho)} \right), \quad W = \text{diag} \left(\frac{1}{4r_\varepsilon(\rho)} \right). \quad (3.6)$$

The Hessian contribution of the regularized kinetic term admits the factorization

$$H_{\text{kin}} = B^T W B + \text{diag} \left(-\frac{q^2 r''_\varepsilon(\rho)}{8r_\varepsilon(\rho)^2} \right). \quad (3.7)$$

Since $r_\varepsilon > 0$ and $r''_\varepsilon \leq 0$ wherever the second derivative exists, both terms in (3.7) are positive semidefinite. The Hessian of the full discrete energy therefore has the decomposition

$$H = H_{\text{kin}} + \delta D_N^T D_N + \beta I + \text{diag}(V p''_\sigma(\rho)). \quad (3.8)$$

Hence H is symmetric positive semidefinite. If $\beta > 0$, it is positive definite. This structure will be used in the Newton refinement in Section 4.2.

For the shifted denominator $r_\varepsilon(s) = s + \varepsilon$, which is used in the Newton refinement below, $r'_\varepsilon = 1$ and $r''_\varepsilon = 0$. The diagonal correction in (3.7) then vanishes. The corresponding matrix-free Hessian action is recorded in Appendix A.

3.2 Positive mass-conservative projection

The constrained optimization requires a projection that preserves nonnegativity and mass. Let S_N denote the trigonometric interpolation space associated with the Fourier grid, and define

$$S_N^+ = \{v_N \in S_N : v_N(x_j) \geq 0, \quad j = 0, \dots, N-1\}. \quad (3.9)$$

where, for $v_N \in S_N$, we use the same discrete norm notation $\|v_N\|_h^2 = h \sum_{j=0}^{N-1} |v_N(x_j)|^2$. For $z_N \in S_N$, consider the positive mass-conservative pseudospectral projection [13]

$$\underset{g_N \in S_N^+}{\text{argmin}} \frac{1}{2} \|g_N - z_N\|_h^2 \quad \text{subject to} \quad \int_{\mathcal{D}_L} g_N \, dx = \int_{\mathcal{D}_L} z_N \, dx. \quad (3.10)$$

The following proposition gives its nodal form.

Proposition 3.2 (Nodal form of the positive mass-conservative projection). *Let $z = (z_j)_{j=0}^{N-1} \in \mathbb{R}^N$, where $z_j = z_N(x_j)$. Then (3.10) is equivalent to*

$$\underset{y \in \mathbb{R}^N}{\text{argmin}} \frac{1}{2} \|y - z\|_h^2$$

subject to

$$y_j \geq 0, \quad h \sum_{j=0}^{N-1} y_j = \int_{\mathcal{D}_L} z_N \, dx.$$

Proof. By the definition of the discrete norm,

$$\|g_N - z_N\|_h^2 = h \sum_{j=0}^{N-1} |g_N(x_j) - z_N(x_j)|^2.$$

Moreover, the trapezoidal rule is exact for trigonometric polynomials, so

$$\int_{\mathcal{D}_L} g_N \, dx = h \sum_{j=0}^{N-1} g_N(x_j).$$

Hence the objective and constraints in (3.10) coincide with those of the nodal problem. $\square$

Proposition 3.2 shows that the pseudospectral projection can be carried out entirely in terms of the nodal values. For the unit-mass constraint used in the optimization, we therefore define the nodal projection onto $\mathcal{C}_N$ by

$$P_{\mathcal{C}_N}(z) = \operatorname{argmin}_{y \in \mathcal{C}_N} \frac{1}{2} \|y - z\|_h^2, \quad z \in \mathbb{R}^N. \quad (3.11)$$

It is given componentwise by

$$[P_{\mathcal{C}_N}(z)]_j = \max\{z_j - \lambda, 0\}, \quad h \sum_{j=0}^{N-1} [P_{\mathcal{C}_N}(z)]_j = 1, \quad (3.12)$$

where λ is uniquely determined by the mass constraint.

4 Two-stage optimization method

In this section, we solve the discrete convex minimization problem using a two-stage strategy. We first apply an accelerated proximal method to reduce the energy efficiently, and then refine the resulting state through the KKT system to reduce the stationarity residual.

The discrete ground-state problem is

$$\min_{\rho \in \mathcal{C}_N} E_{\varepsilon, \sigma, N}(\rho). \quad (4.1)$$

Since the objective and $\mathcal{C}_N$ are convex, every KKT point is a global minimizer.

To assess stationarity in both stages, for a fixed $\tau_r > 0$ define the projected-gradient residual

$$\mathcal{G}_{\tau_r}(\rho) = \frac{1}{\tau_r} [\rho - P_{\mathcal{C}_N}(\rho - \tau_r G_{\varepsilon, \sigma, N}(\rho))]. \quad (4.2)$$

For any feasible ρ , the optimality condition for (4.1) is equivalent to $\mathcal{G}_{\tau_r}(\rho) = 0$. We use $\|\mathcal{G}_{\tau_r}(\rho)\|_h$ as the stationarity measure.

4.1 Potential-proximal FISTA

FISTA is an accelerated proximal-gradient method for composite convex optimization [11]. We apply it to the discrete problem (4.1) by splitting the energy into a smooth part and a proximal part. As shown in (2.9), the potential regularization has curvature $p''_\sigma(0) = 1/\sigma$, which becomes large as σ decreases. We therefore include the regularized potential together with the positivity and mass constraints in the proximal part. Thus we use the splitting

$$E_{\varepsilon, \sigma, N}(\rho) = f_{\varepsilon, N}(\rho) + g_{\sigma, N}(\rho), \quad (4.3)$$

where

$$f_{\varepsilon,N}(\rho) = h \sum_j \left[\frac{(D_N \rho)_j^2}{8r_\varepsilon(\rho_j)} + \frac{\beta}{2} \rho_j^2 + \frac{\delta}{2} (D_N \rho)_j^2 \right] \quad (4.4)$$

is the smooth part, and

$$g_{\sigma,N}(\rho) = h \sum_j V_j p_\sigma(\rho_j) + I_{\mathcal{C}_N}(\rho) \quad (4.5)$$

is the proximal part. Here $I_{\mathcal{C}_N}$ denotes the indicator function of $\mathcal{C}_N$. The gradient of $f_{\varepsilon,N}$ with respect to the discrete inner product (3.1) is

$$\nabla_h f_{\varepsilon,N}(\rho) = \frac{1}{4} D_N^T \left(\frac{D_N \rho}{r_\varepsilon(\rho)} \right) - \frac{1}{8} \frac{(D_N \rho)^2 r'_\varepsilon(\rho)}{r_\varepsilon(\rho)^2} + \beta \rho + \delta D_N^T D_N \rho, \quad (4.6)$$

where the nonlinear operations are understood pointwise.

We use the Chambolle–Dossal variant of FISTA with backtracking [11, 14]. Given two consecutive iterates $\rho^{k-1}, \rho^k \in \mathcal{C}_N$, one FISTA iteration consists of an extrapolation to y^k , a forward step to z^k , and a proximal update to $\rho^{k+1} \in \mathcal{C}_N$.

The extrapolated state is

$$y^k = \rho^k + \theta_k(\rho^k - \rho^{k-1}), \quad (4.7)$$

where θ_k is the Chambolle–Dossal extrapolation parameter. In the computations, $\theta_k = (k-1)/(k+4)$. The extrapolation preserves the mass constraint. Since $f_{\varepsilon,N}$ is defined for nonnegative densities, the extrapolation is accepted only if $y^k \geq 0$. Otherwise, it is restarted by setting

$$y^k = \rho^k.$$

A monotonicity restart also sets $y^k = \rho^k$ when the trial objective exceeds the preceding objective by more than $100u$ times the larger of one and the absolute values of the two objectives, where u is machine precision.

For a trial step size $\tau_k > 0$, the forward step is

$$z^k = y^k - \tau_k \nabla_h f_{\varepsilon,N}(y^k). \quad (4.8)$$

The proximal update is

$$\rho^{k+1} = \operatorname{argmin}_{\rho \in \mathcal{C}_N} \left\{ \frac{1}{2} \left\| \rho - z^k \right\|_h^2 + \tau_k h \sum_j V_j p_\sigma(\rho_j) \right\}. \quad (4.9)$$

The positivity and mass constraints are therefore enforced in the proximal step.

The trial step size is accepted when the standard majorization condition

$$f_{\varepsilon,N}(\rho^{k+1}) \leq f_{\varepsilon,N}(y^k) + \left\langle \nabla_h f_{\varepsilon,N}(y^k), \rho^{k+1} - y^k \right\rangle_h + \frac{1}{2\tau_k} \left\| \rho^{k+1} - y^k \right\|_h^2 \quad (4.10)$$

is satisfied. Otherwise, τ_k is reduced and the forward–proximal step is repeated. The first trial uses $\tau_1 = 1$, and each later iteration starts from the preceding accepted step size. Each backtracking step halves τ_k , with at most 100 trials.

The proximal problem (4.9) can be solved through a single scalar mass multiplier. For a fixed λ , each positive component of ρ^{k+1} satisfies

$$s - z_j^k + \tau_k V_j p'_\sigma(s) + \lambda = 0, \quad s > 0. \quad (4.11)$$

The left-hand side is strictly increasing in s , since $1 + \tau_k V_j p''_\sigma(s) > 0$. Hence the positive solution, when it exists, is unique. The j th component is zero when $z_j^k \leq \lambda$ and otherwise is given by the unique positive solution of (4.11). The multiplier λ is determined by the mass constraint

$$h \sum_j \rho_j^{k+1} = 1.$$

Thus the proximal update consists of independent scalar nodal equations coupled only through the mass constraint. The nodal equations and the scalar mass equation are solved by safeguarded Newton iterations with bisection fallback.

For the unregularized linear potential, $p(s) = s$, (4.9) reduces to

$$\rho^{k+1} = P_{\mathcal{C}_N}(z^k - \tau_k V).$$

Hence the potential-proximal update is a nonlinear extension of the nodal projection (3.11).

Once the energy has stabilized and the projected residual (4.2) is sufficiently small, the iterate is passed to the Newton refinement.

4.2 Newton refinement

After the FISTA stage, we use a Newton refinement to further reduce the stationarity residual (4.2). The corresponding KKT conditions of (4.1) are

$$\begin{aligned} G_{\varepsilon,\sigma,N}(\rho) + \lambda \mathbf{1} - \mu &= 0, & h \mathbf{1}^T \rho &= 1, \\ \rho &\geq 0, & \mu &\geq 0, & \rho_j \mu_j &= 0. \end{aligned} \quad (4.12)$$

When the solution is interior, the inequality multiplier vanishes and (4.12) reduces to

$$G_{\varepsilon,\sigma,N}(\rho) + \lambda \mathbf{1} = 0, \quad h \mathbf{1}^T \rho = 1. \quad (4.13)$$

Starting from the FISTA output, each interior Newton iteration computes a correction from the linearized KKT system and applies a damped update. In the strongly convex regime $\beta > 0$, the Hessian is symmetric positive definite by (3.8). For the current iterate (ρ^k, λ^k) , define

$$r^k = G_{\varepsilon,\sigma,N}(\rho^k) + \lambda^k \mathbf{1}, \quad r_m^k = h \mathbf{1}^T \rho^k - 1$$

as the stationarity and mass residuals, respectively. Let $H^k = H(\rho^k)$ denote the Hessian of $E_{\varepsilon,\sigma,N}$ at ρ^k (3.8). The Newton correction (d^k, d_λ^k) satisfies

$$H^k d^k + d_\lambda^k \mathbf{1} = -r^k, \quad h \mathbf{1}^T d^k = -r_m^k.$$

Rather than solving the indefinite KKT system directly, we use a Schur complement. Let

$$H^k z^k = r^k, \quad H^k w^k = \mathbf{1}.$$

Then

$$d_\lambda^k = \frac{r_m^k - h \mathbf{1}^T z^k}{h \mathbf{1}^T w^k}, \quad d^k = -z^k - d_\lambda^k w^k. \quad (4.14)$$

Thus each interior Newton step requires two linear systems with the same symmetric positive definite Hessian. We solve these systems by the preconditioned conjugate-gradient (PCG) method. With $R_k = \|r^k\|_h$, the relative PCG tolerance is

$$\eta_k = \min \left\{ 10^{-2}, \max \left\{ 10^{-10}, 0.1 R_k^{1/2} \right\} \right\},$$

and each solve is limited to $\min\{500, \max\{100, N\}\}$ iterations.

The correction is applied with a damping factor $\alpha_k \in (0, 1]$:

$$\rho^{k+1} = \rho^k + \alpha_k d^k, \quad \lambda^{k+1} = \lambda^k + \alpha_k d_\lambda^k.$$

The initial damping factor is

$$\alpha_k = \min \left\{ 1, 0.995 \min_{d_j^k < 0} \frac{-\rho_j^k}{d_j^k} \right\},$$

with the second minimum omitted when $d^k \geq 0$. A residual line search halves α_k until $R_{k+1} \leq (1 - 10^{-4}\alpha_k)R_k$ or the projected-gradient target is met. It allows at most 30 backtracking steps and a minimum step of 10^{-12} . If active constraints are encountered, a standard primal-dual active-set treatment can be used [25].

For the PCG solves, we use a sparse variable-coefficient finite-difference approximation of the Fourier pseudospectral Hessian as a symmetric positive definite preconditioner. Its construction is given in Appendix A.

5 Numerical experiments

In this section, we examine the spatial accuracy, regularization effects, and computational performance of the positive-conservative Fourier optimization (PCFO) method, and present representative two-dimensional examples.

Unless otherwise stated, we take $\beta = \delta = 10$, impose the unit-mass constraint, use the shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$, and adopt the potential regularization (2.8). All projected-gradient residuals reported below use $\tau_r = 1$. All two-stage solves use the FISTA-to-Newton switching rule specified in Subsection 5.3. The Newton refinement is limited to 20 iterations. Its configured projected-gradient target is 10^{-12} in the spatial-accuracy, Fourier-resolution, efficiency, and two-dimensional tests, and 10^{-10} in the potential-bias and kinetic-regularization tests. A five-iteration stagnation check also stops the refinement when the relative residual reduction is below 10^{-2} , recording residuals at most 10^{-10} as a numerical floor.

5.1 Spatial accuracy

We first examine the spatial accuracy of the Fourier discretization with fixed regularization parameters. To separate the spatial error from the regularization error, we work on the periodic interval $\mathcal{D}_L = [-L, L]$ with $L = 32$ and fix $\varepsilon = \sigma = 10^{-2}$.

We take the harmonic potential $V_H(x) = x^2/2$. Its periodic extension is continuous but not smoothly periodic, since $V_H'(-L) \neq V_H'(L)$, which limits the Fourier accuracy at high resolution. For the spatial-convergence test, we therefore use

$$V_L^p(x) = (1 - \chi_L(x)) \frac{x^2}{2} + \chi_L(x) \frac{L^2}{2}, \quad (5.1)$$

where χ_L is an even C^∞ cutoff satisfying $\chi_L = 0$ for $|x| \leq 0.75L$ and $\chi_L = 1$ for $|x| \geq 0.90L$. Specifically,

$$\chi_L(x) = S\left(\frac{|x| - 0.75L}{0.15L}\right), \quad S(t) = \begin{cases} 0, & t \leq 0, \\ \frac{\exp(-1/t)}{\exp(-1/t) + \exp(-1/(1-t))}, & 0 < t < 1, \\ 1, & t \geq 1. \end{cases}$$

Thus V_L^p agrees with the harmonic potential in the interior and is constant near the periodic boundary.

Let ρ_N and ρ_{ref} denote the trigonometric interpolants of the numerical solutions on the N -point grid and the finest comparison grid, respectively. Let P_N denote the $L^2(\mathcal{D}_L)$ -orthogonal Fourier projection onto S_N . We define

$$e_{\rho,2}(N) = \|\rho_N - \rho_{\text{ref}}\|_{L^2(\mathcal{D}_L)}, \quad e_{\rho,\infty}(N) = \max_{0 \leq j < N} |\rho_N(x_j) - (P_N \rho_{\text{ref}})(x_j)|. \quad (5.2)$$

The energy error is

$$e_E(N) = |E_{\varepsilon,\sigma,N}(\rho_N) - E_{\varepsilon,\sigma,N_{\text{ref}}}(\rho_{\text{ref}})|. \quad (5.3)$$

Table 5.1: Fourier mesh refinement for $\varepsilon = \sigma = 10^{-2}$ using the harmonic trap with a C^∞ periodic continuation (5.1). The comparison state is computed with $N_{\text{ref}} = 8192$.

N	$e_{\rho,2}$	rate	$e_{\rho,\infty}$	rate	e_E	rate
32	2.519×10^{-2}	—	1.282×10^{-2}	—	5.831×10^{-3}	—
64	3.394×10^{-3}	2.89	1.422×10^{-3}	3.17	2.077×10^{-3}	1.49
128	1.190×10^{-4}	4.83	9.022×10^{-5}	3.98	5.597×10^{-6}	8.54
256	6.227×10^{-7}	7.58	2.526×10^{-7}	8.48	3.770×10^{-9}	10.54
512	6.415×10^{-11}	13.24	4.185×10^{-11}	12.56	4.174×10^{-14}	16.46

Table 5.1 reports the errors relative to the $N_{\text{ref}} = 8192$ comparison state. The state and energy errors decrease rapidly, with increasing empirical orders, until the numerical floor is reached, which is consistent with spectral-type spatial convergence for fixed regularization parameters.

For comparison, we repeat the refinement with the unmodified harmonic potential $V_H = x^2/2$ on the periodic domain. The two calculations are nearly indistinguishable on coarse and moderate meshes. At higher resolutions, the direct harmonic calculation develops an algebraic error floor, with the observed order approaching two, consistent with the lack of periodic smoothness of V_H . This error floor is removed by the C^∞ far-field continuation, as shown in Fig. 5.1.

5.2 Regularization effects

The regularized model contains two parameters associated with the kinetic and potential terms. We first examine the effect of the potential regularization parameter σ at fixed ε , including its influence on Fourier resolution. We then compare different regularizations of the density kinetic term as $\varepsilon \rightarrow 0$.

Potential regularization. We fix $\varepsilon = 10^{-2}$, $L = 32$, and $N = 4096$, and take $\sigma = \varepsilon^p$ with $p = 1, 2, 3, 4$. Let $\rho_{\varepsilon,\sigma}$ denote the minimizer with the regularized potential, and let $\rho_{\varepsilon,0}$ denote the minimizer obtained with the original potential. To quantify the perturbation introduced by p_σ , we define

$$B_E(\sigma) = E_{\varepsilon,0}(\rho_{\varepsilon,\sigma}) - E_{\varepsilon,0}(\rho_{\varepsilon,0}), \quad B_\rho(\sigma) = \|\rho_{\varepsilon,\sigma} - \rho_{\varepsilon,0}\|_{L^2(\mathcal{D}_L)}. \quad (5.4)$$

Here $E_{\varepsilon,0}$ denotes the energy with the same denominator r_ε and the original potential contribution $V\rho$.

Table 5.2 shows that both perturbations decrease approximately linearly with σ over the tested range. The minimum density exhibits the same scaling for small σ , indicating that the positive low-density background introduced by the potential regularization vanishes together with σ .

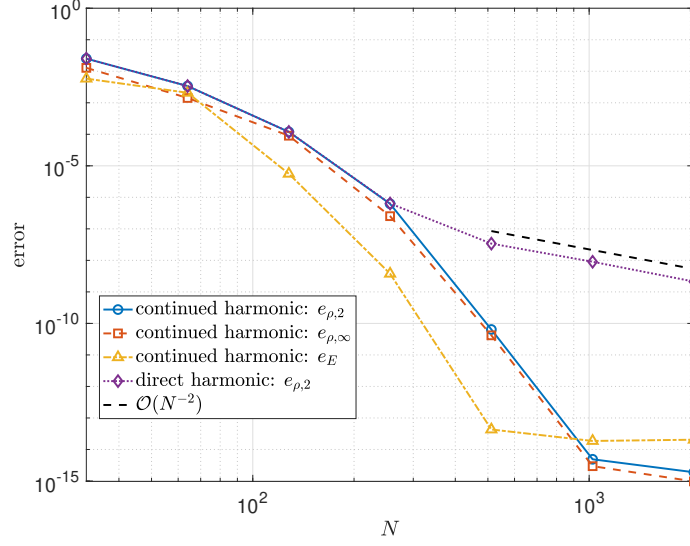


Figure 5.1: Fourier spatial convergence with fixed regularization parameters. “Continued harmonic” refers to the C^∞ far-field continuation (5.1), and “direct harmonic” to $V_H(x) = x^2/2$ on the periodic computational domain.

Table 5.2: Effect of the potential regularization scale at $\varepsilon = 10^{-2}$, $L = 32$, and $N = 4096$. The quantities $B_E(\sigma)$ and $B_\rho(\sigma)$ are defined in (5.4).

σ	$B_E(\sigma)$	$B_\rho(\sigma)$	$\min \rho$
10^{-2}	1.776×10^0	2.548×10^{-2}	6.449×10^{-5}
10^{-4}	1.849×10^{-2}	2.900×10^{-4}	6.740×10^{-7}
10^{-6}	1.849×10^{-4}	2.719×10^{-6}	6.743×10^{-9}
10^{-8}	1.849×10^{-6}	1.993×10^{-8}	6.744×10^{-11}

To examine the transition introduced by the potential regularization, we consider

$$p'_\sigma(\rho_{\varepsilon,\sigma}(x)) = \frac{\rho_{\varepsilon,\sigma}(x)}{\sqrt{\rho_{\varepsilon,\sigma}(x)^2 + \sigma^2}}, \quad (5.5)$$

which measures the local slope of the regularized potential relative to the original linear potential. It is close to one when $\rho_{\varepsilon,\sigma} \gg \sigma$ and close to zero when $\rho_{\varepsilon,\sigma} \ll \sigma$.

Figure 5.2 shows that the transition between these two regimes becomes sharper as σ decreases. Accordingly, the modification introduced by the potential regularization becomes more localized near the low-density region. The resulting smaller spatial scale requires more Fourier modes to resolve accurately.

To examine this effect on Fourier resolution, we repeat the mesh refinement for $\sigma = 0, 10^{-2}, 10^{-4}$, and 10^{-6} , while keeping $\varepsilon = 10^{-2}$ and $L = 32$. Here $\sigma = 0$ corresponds to the original linear potential without potential regularization. We use the same C^∞ -continued harmonic potential (5.1) and the $N_{\text{ref}} = 65536$ solution as the reference.

Figure 5.3 shows the resulting energy and state errors. For $\sigma = 10^{-2}$, both errors rapidly enter the spectral-type regime and reach their numerical floors. As σ decreases, a longer pre-asymptotic range appears before the rapid decay becomes visible. In contrast, for $\sigma = 0$, the nonzero slope of the linear potential at zero density can activate the positivity constraint and reduce the regularity

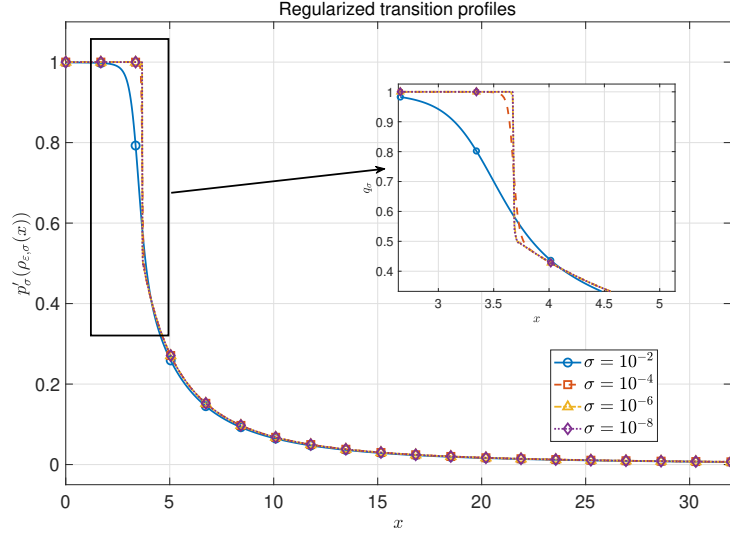


Figure 5.2: Profiles of $p'_\sigma(\rho_{\varepsilon,\sigma}(x))$ (5.5) for different potential-regularization scales. The transition becomes sharper as σ decreases, and the inset enlarges the transition region. All computations use $\varepsilon = 10^{-2}$, $L = 32$, and $N = 4096$.

of the density. Consistently, spectral-type decay is not observed over the tested resolutions, and the errors decrease much more slowly. This reveals a bias–resolution tradeoff in which decreasing σ reduces the regularization error but requires a finer Fourier mesh to maintain the same spatial accuracy.

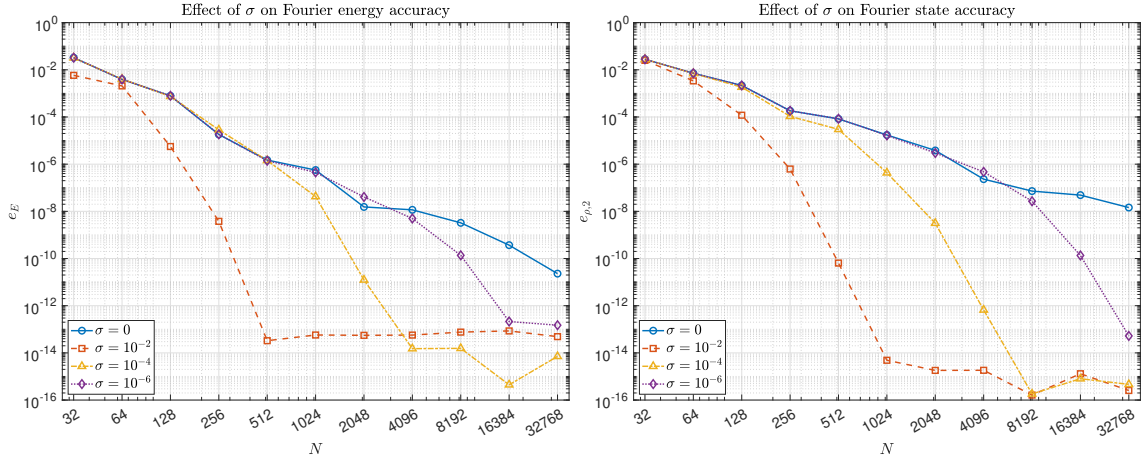


Figure 5.3: Effect of the potential regularization scale on Fourier accuracy. The left panel shows the energy error and the right panel shows the L^2 state error. Each curve refines a fixed- σ problem with $\varepsilon = 10^{-2}$, $L = 32$, and the harmonic-core C^∞ far-field continuation, relative to the $N_{\text{ref}} = 65536$ comparison state.

Regularization of the kinetic term. We next show the effect of the kinetic regularization by fixing the potential regularization and the spatial discretization. We compare the shifted denominator (2.6) with the piecewise C^1 and C^2 denominators in (2.7). Throughout this test, we take

$\sigma = 10^{-2}$, $L = 32$, and $N = 512$, and use the same C^∞ -continued harmonic potential. The tested values are $\varepsilon = 10^{-1}, 5 \times 10^{-2}, 2 \times 10^{-2}, 10^{-2}, 5 \times 10^{-3}, 2 \times 10^{-3}, 10^{-3}$.

A common reference state ρ_{ref} is computed with the shifted denominator at $\varepsilon_{\text{ref}} = 10^{-5}$. To compare the three regularizations on the same basis, we measure the state error and evaluate the energy error with the common unregularized kinetic denominator $r_0(\rho) = \rho$. We define

$$e_\rho(\varepsilon) = \|\rho_\varepsilon - \rho_{\text{ref}}\|_{L^2(\mathcal{D}_L)}, \quad e_E(\varepsilon) = |E_{0,\sigma}(\rho_\varepsilon) - E_{0,\sigma}(\rho_{\text{ref}})|. \quad (5.6)$$

Here $E_{0,\sigma}$ denotes the energy with the original kinetic denominator $r_0(s) = s$ and the fixed potential regularization p_σ .

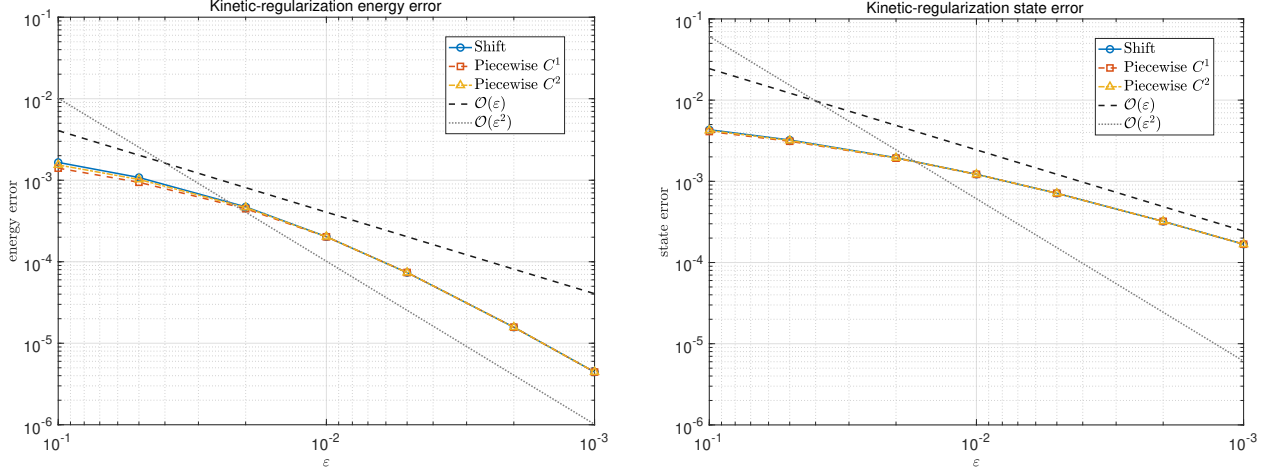


Figure 5.4: Effect of the kinetic regularization. The shifted denominator (2.6) and the piecewise C^1 and C^2 denominators (2.7) are compared using the errors defined in (5.6). The left panel shows the energy error $e_E(\varepsilon)$ and the right panel the L^2 state error $e_\rho(\varepsilon)$. The dashed and dotted lines indicate first- and second-order slopes, respectively. All computations use $\sigma = 10^{-2}$, $L = 32$, and $N = 512$.

Figure 5.4 shows that the three regularizations have essentially the same asymptotic behavior as ε decreases. The common-functional energy error approaches second order whereas the state error is approximately first order. The piecewise C^1 denominator gives a modest reduction in error for the larger values of ε , but neither piecewise regularization shows a clear asymptotic rate advantage over the shifted denominator. This supports using the simpler shifted denominator in the solver experiments.

5.3 Efficiency tests

We next examine the computational performance of the two-stage optimization strategy. Throughout this subsection, we use the shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$. We first examine the respective roles of FISTA and the Newton refinement, and then examine how the two regularization scales affect the computational cost.

FISTA and Newton refinement. We consider the representative case $\varepsilon = 10^{-2}$, $\sigma = 10^{-4}$, $L = 32$, and $N = 4096$. All runs start from the projected Gaussian

$$\rho^0 = P_{C_N} \left((e^{-x_j^2} / \sqrt{\pi})_{j=0}^{N-1} \right).$$

For $k \geq 200$, define the relative energy-window span

$$\Delta_k^{(50)} = \frac{\max_{k-49 \leq \ell \leq k} E_{\varepsilon, \sigma, N}(\rho^\ell) - \min_{k-49 \leq \ell \leq k} E_{\varepsilon, \sigma, N}(\rho^\ell)}{\max\{1, |E_{\varepsilon, \sigma, N}(\rho^k)|\}}.$$

The FISTA stage is switched to the Newton refinement when $\Delta_k^{(50)} \leq 10^{-12}$ and $\|\mathcal{G}_1(\rho^k)\|_h \leq 10^{-5}$ for two consecutive iterations. Starting from the same switching state, one branch continues with FISTA, while the other proceeds with the Newton refinement. The two potential-proximal runs therefore share the same FISTA history up to the switch.

For comparison, we also consider a smooth-potential FISTA, in which the regularized potential is included in the smooth part instead of the proximal part. It minimizes the same discrete energy and uses the same FISTA backtracking rule.

Since FISTA and Newton steps have different computational costs, Fig. 5.5 reports the convergence histories against cumulative wall-clock time. The two branches share the same FISTA time history up to the switching point. Thereafter, the additional time of each branch is measured from the common switching time. The left panel shows the energy error relative to the final two-stage solution, while the right panel shows the projected-gradient residual. As shown in Fig. 5.5, FISTA rapidly reduces the energy and produces an accurate approximation of the ground state, but continued FISTA decreases the projected-gradient residual only slowly. Starting from the same switching state, the Newton refinement reduces the residual much more rapidly.

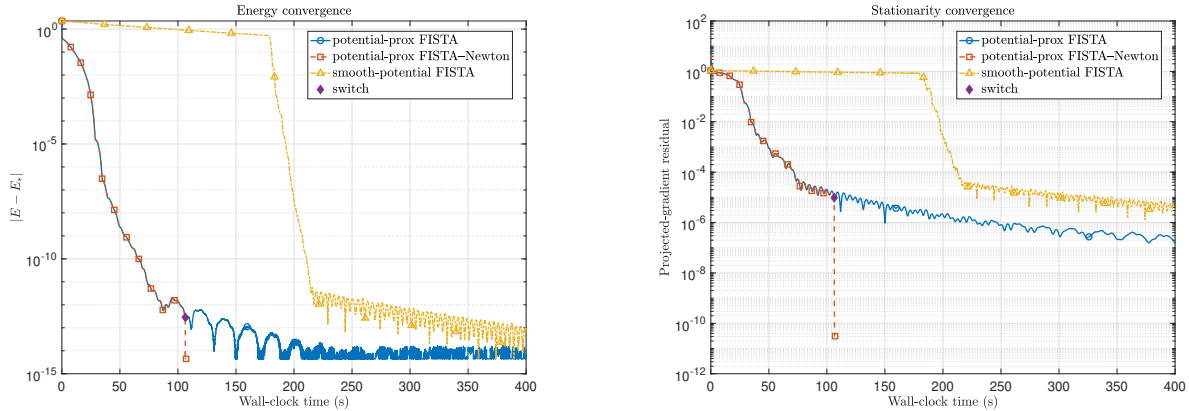


Figure 5.5: Convergence histories for continued FISTA and the two-stage FISTA–Newton method with $\varepsilon = 10^{-2}$, $\sigma = 10^{-4}$, $L = 32$, and $N = 4096$. The left panel shows the energy error relative to the final two-stage solution, and the right panel shows the projected-gradient residual $\|\mathcal{G}_1(\rho^k)\|_h$. Here “potential-proximal FISTA” and “smooth-potential FISTA” refer to placing the regularized potential in the proximal and smooth parts of the splitting, respectively.

Dependence on the regularization scales. We next examine how the computational cost changes as the regularization parameters decrease. To isolate the two effects, we first vary ε at fixed $\sigma = 10^{-2}$ and then vary σ at fixed $\varepsilon = 10^{-2}$. In all cases, $r_\varepsilon(\rho) = \rho + \varepsilon$.

All runs use the same Fourier resolution $N = 4096$, initial density, and solver parameters. Thus the comparison isolates the effect of the regularization scales at a fixed Fourier resolution. The results are summarized in Table 5.3. The reported wall-clock time includes both the FISTA and Newton stages. The FISTA stage is limited to 20000 iterations. At this limit, the Newton refinement is entered only if $\|\mathcal{G}_1(\rho)\|_h \leq 10^{-4}$. The asterisk in Table 5.3 marks the run that reached this limit.

Table 5.3: Computational cost of the two-stage solver for different regularization scales. The shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$ is used in all cases, with the same initialization, Fourier resolution $N = 4096$, and solver parameters. “FISTA” and “Newton” denote the iteration counts in the two stages. The last column reports the final projected-gradient residual $\|\mathcal{G}_1(\rho)\|_h$ (4.2). The asterisk indicates that the FISTA stage reached the prescribed iteration limit.

ε	σ	FISTA	Newton	time (s)	$\ \mathcal{G}_1(\rho)\ _h$
10^{-2}	10^{-2}	3422	8	93.50	5.04×10^{-11}
10^{-3}	10^{-2}	9042	10	201.64	2.54×10^{-10}
10^{-4}	10^{-2}	20000*	14	519.77	1.15×10^{-9}
10^{-2}	10^{-3}	2992	10	99.28	3.93×10^{-11}
10^{-2}	10^{-4}	3025	18	107.02	3.11×10^{-11}

Table 5.3 shows that decreasing ε at fixed $\sigma = 10^{-2}$ substantially increases the FISTA iteration count and the total solution time. The smallest ε case reaches the prescribed FISTA iteration limit and has a somewhat larger final residual. This increase is consistent with the larger coefficients involving $1/r_\varepsilon(\rho)$ in the kinetic Hessian, particularly in low-density regions.

In contrast, decreasing σ at fixed $\varepsilon = 10^{-2}$ produces only a modest change in the computational cost over the tested range. Although a smaller σ sharpens the potential transition, the potential-proximal splitting keeps the large curvature $p''_\sigma(0) = 1/\sigma$ out of the smooth FISTA step.

5.4 Two-dimensional examples

We conclude with two examples in two dimensions. In both cases, we use the shifted denominator $r_\varepsilon(\rho) = \rho + \varepsilon$, the potential regularization p_σ , and

$$\beta = \delta = 10, \quad \varepsilon = 10^{-2}, \quad \sigma = 10^{-4}.$$

The computations are performed on the periodic domain $[-16, 16]^2$ with $N_x = N_y = 512$ Fourier collocation points. The continuation (5.1) is applied separately in the x - and y -directions. Thus the numerical harmonic potential is $\gamma_x^2 V_L^p(x) + \gamma_y^2 V_L^p(y)$ and agrees with the stated trap on $|x|, |y| \leq 0.75L$. Both computations start from $\rho_{ij}^0 = c \exp[-(x_j^2 + y_i^2)/2]$, where c enforces unit mass, and use the switching and Newton stopping parameters stated in Subsection 5.3.

We first consider the anisotropic harmonic trap

$$V_{\text{ah}}(x, y) = \frac{1}{2} (\gamma_x^2 x^2 + \gamma_y^2 y^2), \quad \gamma_x = 1, \quad \gamma_y = 2. \quad (5.7)$$

The resulting ground-state density is shown in the left panel of Fig. 5.6. The stronger confinement in the y -direction gives a density profile that is narrower in that direction.

For the second example, we add a square optical lattice,

$$V_{\text{ol}}(x, y) = V_{\text{ah}}(x, y) + V_0 [\sin^2(kx) + \sin^2(ky)], \quad (5.8)$$

with $V_0 = 5$ and $k = \pi/2$. The resulting density is shown in the right panel of Fig. 5.6. The optical lattice introduces multiple local wells and produces the corresponding modulation of the ground-state density.

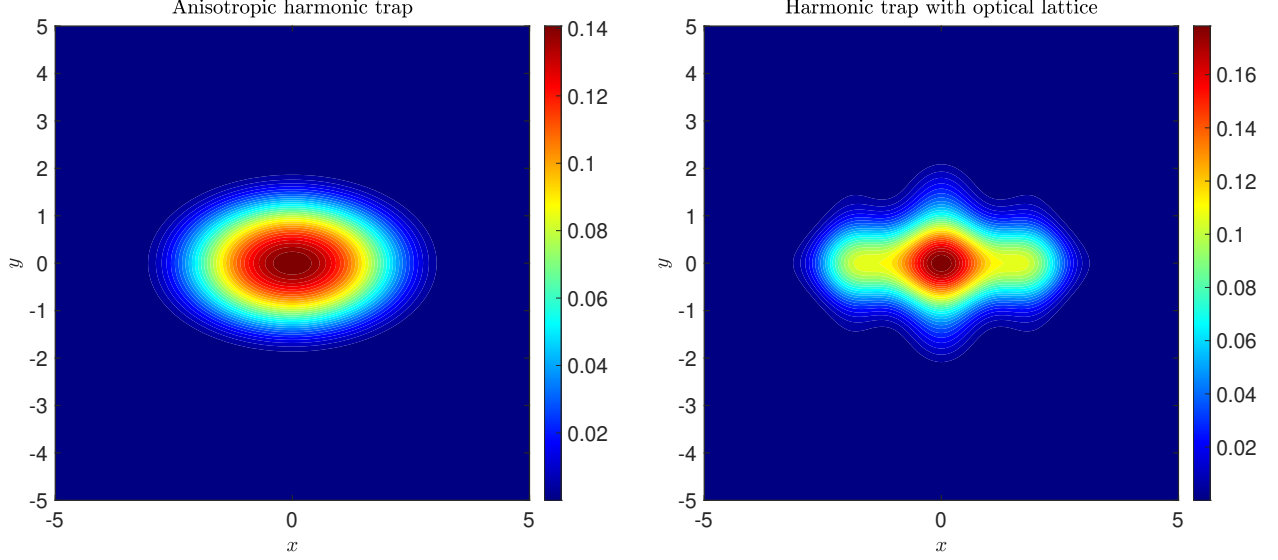


Figure 5.6: Two-dimensional ground-state densities computed on $[-16, 16]^2$ with $N_x = N_y = 512$, with the central region shown. The left panel corresponds to the anisotropic harmonic trap (5.7), while the right panel includes the square optical lattice (5.8). Both computations use $\varepsilon = 10^{-2}$, $\sigma = 10^{-4}$, and $\beta = \delta = 10$.

6 Conclusion

We have developed a positive-conservative Fourier optimization method for computing ground states of Bose–Einstein condensates with higher-order interactions in the density formulation. The kinetic and potential terms are regularized while preserving the convex structure of the density energy. On the computational domain, the regularized energies Γ -converge to the original density energy as the regularization parameters vanish.

The numerical results show spectral-type convergence of the Fourier discretization when the regularization parameters are fixed. They also show that decreasing the potential regularization parameter reduces the regularization error but requires a finer Fourier resolution to resolve the sharper low-density transition. In the two-stage optimization, FISTA provides the main energy reduction, while the subsequent Newton refinement further reduces the stationarity residual to high accuracy.

A joint analysis of domain truncation, Fourier discretization, and regularization remains for future work.

Acknowledgements This work was partially supported by the National Natural Science Foundation of China under Grants 12201436 (X. Ruan) and 12501547 (B. Lin).

A Implementation of the Newton refinement

For the Newton refinement, let

$$a = r_\varepsilon(\rho), \quad a' = r'_\varepsilon(\rho), \quad a'' = r''_\varepsilon(\rho), \quad q = D_N \rho, \quad \dot{q} = D_N \eta.$$

The products and quotients in the following formula are understood componentwise. The Hessian action of the regularized kinetic term is

$$H_{\text{kin}}(\rho)[\eta] = \frac{1}{4}D_N^T \left(\dot{q} - \frac{q a'}{a^2} \eta \right) - \frac{1}{8} \left[2q\dot{q} \frac{a'}{a^2} + q^2 \left(\frac{a''}{a^2} - \frac{2(a')^2}{a^3} \right) \eta \right]. \quad (\text{A.1})$$

The full Hessian action is

$$H(\rho)[\eta] = H_{\text{kin}}(\rho)[\eta] + V p_\sigma''(\rho) \eta + \beta \eta + \delta D_N^T D_N \eta. \quad (\text{A.2})$$

For the PCG preconditioner, let D_{FD} denote the periodic forward-difference operator and define

$$B_{\text{FD}} = D_{\text{FD}} - \text{diag} \left(\frac{q a'}{a} \right), \quad W = \text{diag} \left(\frac{1}{4a} \right), \quad R_{\text{kin}} = \text{diag} \left(-\frac{q^2 a''}{8a^2} \right).$$

We use the sparse matrix

$$P_{\text{FD}} = B_{\text{FD}}^T W B_{\text{FD}} + R_{\text{kin}} + \delta D_{\text{FD}}^T D_{\text{FD}} + \text{diag}(\beta + V p_\sigma''(\rho)) \quad (\text{A.3})$$

as the PCG preconditioner. In the strongly convex regime, P_{FD} is symmetric positive definite.

References

- [1] R. Altmann, P. Henning, and D. Peterseim, The J-method for the Gross–Pitaevskii eigenvalue problem, *Numer. Math.*, 148 (2021), pp. 575–610.
- [2] M. H. Anderson, J. R. Ensher, M. R. Matthews, C. E. Wieman, and E. A. Cornell, Observation of Bose–Einstein condensation in a dilute atomic vapor, *Science*, 269 (1995), pp. 198–201.
- [3] X. Antoine and R. Duboscq, Robust and efficient preconditioned Krylov spectral solvers for computing the ground states of fast rotating and strongly interacting Bose–Einstein condensates, *J. Comput. Phys.*, 258 (2014), pp. 509–523.
- [4] X. Antoine, A. Levitt, and Q. Tang, Efficient spectral computation of the stationary states of rotating Bose–Einstein condensates by preconditioned nonlinear conjugate gradient methods, *J. Comput. Phys.*, 343 (2017), pp. 92–109.
- [5] W. Bao and Y. Cai, Mathematical theory and numerical methods for Bose–Einstein condensation, *Kinet. Relat. Models*, 6 (2013), pp. 1–135.
- [6] W. Bao, Y. Cai, and X. Ruan, Ground states of Bose–Einstein condensates with higher order interaction, *Physica D*, 386–387 (2019), pp. 38–48.
- [7] W. Bao, I.-L. Chern, and F. Y. Lim, Efficient and spectrally accurate numerical methods for computing ground and first excited states in Bose–Einstein condensates, *J. Comput. Phys.*, 219 (2006), pp. 836–854.
- [8] W. Bao and Q. Du, Computing the ground state solution of Bose–Einstein condensates by a normalized gradient flow, *SIAM J. Sci. Comput.*, 25 (2004), pp. 1674–1697.
- [9] W. Bao and X. Ruan, Computing ground states of Bose–Einstein condensates with higher order interaction via a regularized density function formulation, *SIAM J. Sci. Comput.*, 41 (2019), pp. B1284–B1309.

- [10] W. Bao and W. Tang, Ground-state solution of Bose–Einstein condensate by directly minimizing the energy functional, *J. Comput. Phys.*, 187 (2003), pp. 230–254.
- [11] A. Beck and M. Teboulle, A fast iterative shrinkage-thresholding algorithm for linear inverse problems, *SIAM J. Imaging Sci.*, 2 (2009), pp. 183–202.
- [12] A. Braides, *Gamma-Convergence for Beginners*, Oxford University Press, Oxford, 2002.
- [13] Z. Cai, B. Lin and M. Lin, A positive and moment-preserving Fourier spectral method, *SIAM J. Numer. Anal.*, 62 (2024), pp. 273–294.
- [14] A. Chambolle and C. Dossal, On the convergence of the iterates of the “fast iterative shrinkage/thresholding algorithm”, *J. Optim. Theory Appl.*, 166 (2015), pp. 968–982.
- [15] Z. Chen, J. Lu, Y. Lu, and X. Zhang, On the convergence of Sobolev gradient flow for the Gross–Pitaevskii eigenvalue problem, *SIAM J. Numer. Anal.*, 62 (2024), pp. 667–691.
- [16] C. Chin, R. Grimm, P. Julienne, and E. Tiesinga, Feshbach resonances in ultracold gases, *Rev. Mod. Phys.*, 82 (2010), pp. 1225–1286.
- [17] A. Collin, P. Massignan, and C. J. Pethick, Energy-dependent effective interactions for dilute many-body systems, *Phys. Rev. A*, 75 (2007), 013615.
- [18] I. Danaila and P. Kazemi, A new Sobolev gradient method for direct minimization of the Gross–Pitaevskii energy with rotation, *SIAM J. Sci. Comput.*, 32 (2010), pp. 2447–2467.
- [19] I. Danaila and B. Protas, Computation of ground states of the Gross–Pitaevskii functional via Riemannian optimization, *SIAM J. Sci. Comput.*, 39 (2017), pp. B1102–B1129.
- [20] H. Fu, Y. Wang, and B. Gao, Beyond the Fermi pseudopotential: A modified Gross–Pitaevskii equation, *Phys. Rev. A*, 67 (2003), 053612.
- [21] P. Henning and D. Peterseim, Sobolev gradient flow for the Gross–Pitaevskii eigenvalue problem: Global convergence and computational efficiency, *SIAM J. Numer. Anal.*, 58 (2020), pp. 1744–1772.
- [22] X. Ruan, A normalized gradient flow method with attractive–repulsive splitting for computing ground states of Bose–Einstein condensates with higher-order interaction, *J. Comput. Phys.*, 367 (2018), pp. 374–390.
- [23] X. Ruan, Y. Cai, and W. Bao, Mean-field regime and Thomas–Fermi approximations of trapped Bose–Einstein condensates with higher-order interactions in one and two dimensions, *J. Phys. B*, 49 (2016), 125304.
- [24] T. Tian, Y. Cai, X. Wu, and Z. Wen, Ground states of spin- F Bose–Einstein condensates, *SIAM J. Sci. Comput.*, 42 (2020), pp. B983–B1013.
- [25] M. Ulbrich, *Semismooth Newton Methods for Variational Inequalities and Constrained Optimization Problems in Function Spaces*, SIAM, Philadelphia, 2011.
- [26] H. Veksler, S. Fishman, and W. Ketterle, Simple model for interactions and corrections to the Gross–Pitaevskii equation, *Phys. Rev. A*, 90 (2014), 023620.

- [27] X. Wu, Z. Wen, and W. Bao, A regularized Newton method for computing ground states of Bose–Einstein condensates, *J. Sci. Comput.*, 73 (2017), pp. 303–329.
- [28] N. T. Zinner and M. Thøgersen, Stability of a Bose–Einstein condensate with higher-order interactions near a Feshbach resonance, *Phys. Rev. A*, 80 (2009), 023607.